

Theoretical physicist J. Robert Oppenheimer's widespread intellect encompassed molecular physics, quantum theory, and black holes, as well as the mystical reaches of Hinduism. He is most renowned as the mastermind of the Manhattan Project, the US-led atomic weapons program.

MILESTONES

PHD IN PHYSICS

Awarded his doctorate in 1927 from University of Göttingen, Germany, supervised by Max Born.

WEAPONS RESEARCH

Begins mammoth task of assembling scientific minds to work on the Manhattan Project in 1942.

POSTWAR ACADEMIC

Appointed director of the Institute for Advanced Studies in Princeton, New Jersey, in 1947.

REGAINS REPUTATION

Loses government's trust during the Cold War; later reattains it and wins the 1963 Enrico Fermi Prize.

Born into a Jewish family in New York City in 1904, J. Robert Oppenheimer was fascinated by minerals, physics, and chemistry as a child. Aged 12, he was invited to lecture at New York's Mineralogical Club, which had assumed the author of the high-level letters it received must be an adult. Oppenheimer graduated at the top of his high school class and enrolled at Harvard in 1922. There he studied Latin, Greek, Eastern philosophy, poetry, and the sciences and majored in chemistry in 1925. But theoretical physics was his real passion, and he went to Cambridge University to study the subject as a postgraduate.

Europe's leading physicists were then developing quantum theory, and Oppenheimer was invited to the University of Göttingen, Germany, to work with eminent physicist Max Born. Together, in 1927, they produced the Born–Oppenheimer approximation, which explained, as Oppenheimer put it, “why molecules are molecules.” Extending quantum mechanics beyond single atoms to describe the energy of chemical compounds,

“Science is not everything, **but science is very beautiful.**”

J. Robert Oppenheimer, 1966

The Trinity test site in New Mexico was the scene of the world's first nuclear explosion. Oppenheimer (center left) likened the terrible force of the blast to words from the Sanskrit scripture the Bhagavad Gita: “I am become Death, the destroyer of worlds.”

